

Early Blight Management for Organic Tomato Production

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Introduction

Early blight is caused by *Alternaria solani* and *A. tomatophila*, which survive between crops on infected crop residues and on solanaceous host weeds. These fungi can also be carried on tomato seed. Early blight is common on tomatoes and potatoes, and it occasionally infects eggplants and peppers. It causes direct losses by the infection of fruits and indirect losses through leaf lesions, which reduce plant vigor. The disease is favored by warm temperatures and extended periods of leaf wetness from frequent rain, overhead irrigation, or dew. The fungal spores can be spread by wind and rain, irrigation, insects, workers, and on tools and equipment. Once the primary infections have occurred, they become the most important source of new spore production and are responsible for rapid disease spread. Early blight can develop quickly mid- to late season and is more severe when plants are stressed by poor nutrition, drought, other diseases, or pests.

Symptoms

Early blight first appears as small brown-to-black lesions on older foliage. The tissue surrounding the primary lesions may become bright yellow, and when lesions are numerous, entire leaves may become chlorotic. As the lesions enlarge, they often develop concentric rings giving them a bull's-eye or target-spot appearance (Fig. 1). When conditions are favorable for disease development, lesions can become numerous and plants defoliate, reducing both fruit quantity and quality. Fruit can become infected either in the green or ripe stage through the stem attachment. Lesions can become quite large, involve the whole fruit, and have characteristic concentric rings. Infected fruit often drops, and losses of immature fruit may occur. Fruit on defoliated plants are also subject to sunscald. Stems and petioles affected by early blight have elliptical concentric lesions, which severely weaken the plant (Fig. 2). Lesions at the base of emerging seedlings can cause collar rot. If this arises consecutively on many seedlings, it may indicate contamination of tomato seeds or soil used for planting.

Figure 1. Early blight on tomato leaf. Figure credit: Fulya Baysal-Gurel, The Ohio State University

Figure 2. Early blight on tomato stems. Figure credit: Thomas A Zitter, Cornell University



Management

Seed selection and treatments

The planting of pathogen-free organically produced tomato seed is an important first step in managing early blight disease. Further information on organic tomato seed selection can be found in the related eOrganic article [Sourcing Certified Organic Seed and the National Organic Program Regulations](#).

Fungicidal seed treatments are not an option for organic growers; however, there are some seed treatments—such as hot water sanitation or [National Organic Program \(NOP\)](#)-compliant protectants—that can be used by organic farmers to eradicate some pathogens from seed. Hot water seed treatment at 122°F for 25 minutes is

recommended to control early blight on tomato seed. Chlorine seed treatment is not an acceptable treatment. Growers should always check with their certifying agency prior to using any seed treatment. For more information on organic seed treatments, see the eOrganic article [Organic Seed Treatments and Coatings](#).



Variety selection

A number of tomato varieties have been developed that are partially resistant to early blight, such as the Mountain series—including Mountain Pride, Mountain Supreme, Mountain Gold, Mountain Fresh, and Mountain Belle. Such varieties are sometimes inaccurately described as "tolerant" to early blight. Partially resistant varieties are not immune to a particular disease; in the case of partial resistance to early blight, infected plants may develop lesions or defoliate to some degree, often depending on environmental conditions, plant stress, and other factors. Further information on disease-resistant tomato varieties can be found in databases provided by Cornell University at <http://vegetablemdonline.ppath.cornell.edu/>.

Propagation and Nursery Management

The planting of pathogen-free organically produced tomato transplants is essential in managing early blight disease. Seeds should be sown in a [pathogen-free planting mix that meets organic standards](#). Field soil may contain pathogens, weed seeds, and insect pests, so it is not recommended for seedling production. Flats with larger cells allow greater air movement between seedlings, which promote rapid drying of foliage and discourages disease development. Mulch (black plastic, straw, and newspaper, for example) helps to protect the plant from inoculum splashing from the soil onto lower leaves.

Crop Rotation

Early blight is a soilborne disease, so rotation can be a good management tool. A good practice is to treat members of the same plant family as a group and rotate based on groups rather than individual crops. Solanaceous crops include tomatoes, potatoes, peppers, chilies, eggplants, and tobacco. Using a three or four year crop rotation with non-solanaceous crops will allow infested plant debris to decompose in the soil. Rotations with small grains, corn, or legumes are preferable.

Organic amendments

[Good quality compost](#) improves soil structure and its ability to hold water and nutrients; it also supports microorganisms that contribute to biological control. Our research has shown that early blight severity was less in tomato plants grown in compost-amended soil in the high tunnel than in non-amended soil; furthermore, incorporating the amendments into soil increased the total and marketable yield (Baysal-Gurel et al., 2009).

Sanitation

Early blight survives between crops in or on the residue from diseased plants, so it is important to remove diseased plants or destroy them immediately after harvest. Alternatively, bury diseased crop debris by deep-plowing to reduce spore levels available for infection of new plants. Solanaceous weeds, such as jimsonweed, horse nettle, ground-cherry, and the numerous nightshades, should be eliminated as they may harbor pathogen inoculum. Volunteer potatoes and tomatoes can also be a source of inoculum for early blight. Frequent disinfection of pruning tools should be disinfected frequently during use to help prevent the spread of spores. Stakes and cages can be disinfected each season before use with an approved product, such as ethanol, hydrogen peroxide, or peracetic acid. Disinfection with sodium hypochlorite (bleach) at 0.5% is effective, but must be followed by rinsing.

Materials

Early detection is important in disease management, so it is important to scout plants regularly. Disease forecasting is another important practice used to predict the probability of disease incidence. Weather monitoring instruments are placed in the field to collect data on canopy temperature, leaf wetness periods, and other factors that affect the likelihood of disease occurrence. The data collected from these monitoring stations are used to time fungicidal sprays for their optimum effect, generally resulting in fewer spray applications each growing season. If the uses of preventive and cultural methods for disease control are insufficient to manage early blight, [National Organic Program \(NOP\)](#)-compliant inputs can be applied to reduce disease spread. NOTE: Before applying any pest control product, be sure to (1) read the label to be sure that the product is labeled for the crop you intend to

apply it to and the disease you intend to control, (2) read and understand the safety precautions and application restrictions, and (3) make sure that the brand name product is listed in your [Organic System Plan](#) and is approved by your certifier. For more information on how to determine whether a disease management product can be used on your farm, see the eOrganic article [Can I Use this Product for Disease Management on my Organic Farm ?](#)

Copper products are considered synthetic and allowed with restrictions. Fixed copper products, hydrogen dioxide (=hydrogen peroxide), and potassium bicarbonate can be used against early blight. However, over-application of copper products can lead to copper accumulation in the soil, contamination of run-off water, and subsequent toxicity to non-target organisms. Biorational products represent an important option for the management of plant diseases. Research has shown that *A. solani*-inoculated tomato plants treated with compost extract, prepared in a ratio of 1:5 compost:water (v/v), showed a significant reduction in disease index as compared with the untreated inoculated plants (Tsrer, 1999). Other research has shown that the efficacy of compost tea was improved when combined with the biofungicides Serenade Max™ (*Bacillus subtilis*) and Sonata™ (*Bacillus pumilis*). These results indicate that the use of compost tea for control of tomato early blight disease may be of some benefit to greenhouse tomato growers, and to growers of organic field tomatoes who are limited in their disease management options (Kouyoumjian, 2007). Garlic- and neem oils and seaweed extract have also been shown to be effective in reducing the severity of early blight disease on tomato compared to untreated controls in another research project (Wszelaki and Miller, 2005). For more information about microbial biopesticides, see the related eOrganic article [Biopesticides for Plant Disease Management in Organic Farming](#).

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